



Fullstack Developer Technical Test

Susi Air · Pilot App · Nuxt 3 + NestJS

This technical test is part of our recruitment process for the Fullstack Developer position at Susi Air. Please read the brief carefully before you start. If anything is unclear, just contact us it.admin@susiair.com.

Project Context

You will build a small piece of the Susi Air Pilot App. It is a mobile web app that pilots use to see their schedule, log flight hours, and track how close they are to their duty limits.

Both the frontend and the backend are part of this test.

Tech Stack

Frontend

- **Nuxt 3** with Composition API and `<script setup>`
- **Pinia** for state management
- **SCSS** for styling

Backend

- **NestJS** on Node.js
- **TypeScript** on the backend. Preferred on the frontend.
- **Data source:** seed the service from the provided JSON files at startup. No database is required. If you want to use SQLite with Prisma or TypeORM, that is fine, but it is not expected.

Scope

Backend: REST API

Build a NestJS service that exposes the following endpoints. Load the provided JSON files as the data source when the service boots.

Method	Path	Purpose
POST	/auth/login	Accept { username, password }. Return a token. Hardcode one pilot account.
GET	/pilot/me	Return pilot profile: name, total flight hours, avatar URL.
GET	/flight-hours?from=YYYY-MM-DD&to=YYYY-MM-DD	Return daily flight hours in the range.
GET	/flight-hours/summary?range=1w 1m 3m 6m 1y	Return the rolling sum series for the chart. The rolling sum runs on the server.

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GET	/documents	Return the pilot's documents with expiry dates.
GET	/schedules?year=YYYY&month=MM	Return schedule entries for one month.

Requirements:

- Proper module, controller, service, and DTO separation.
- Input validation on endpoints that take parameters.
- A global exception filter and a consistent error response shape.
- An auth guard on every endpoint except /auth/login. The guard should check the token issued by /auth/login.
- The rolling sum used by /flight-hours/summary must run on the server, not on the client.
- Username : johndoe
Password : susiairtest

Frontend: Three Screens

Build the following three screens and wire them to your backend. Do not keep mock data in the frontend; fetch everything from the API.

1. Sign In page

- Pilot login form with username and password.
- Show a clear message when the API returns a bad credentials error.

2. Home page

- Header: greeting, pilot name, total flight hours, and avatar.
- Hours to Limit section: four cards plus a rolling sum chart with a 1w, 1m, 3m, 6m, 1y toggle. Full spec below. Chart data comes from /flight-hours/summary.
- My Documents list. Each item shows an expiry badge colored by urgency (green = safe, amber = soon, red = expired). The badge state comes from the API.
- Bottom navigation: Home, Schedule, Logbook, More.

3. Schedule page

- Monthly calendar with previous and next month buttons. Every change calls /schedules?year=YYYY&month=MM.
- Days are colored by duty type (DUTY, RL, SCK, TR, TX, ADM, FER, MED, REC, UL) using base_color from the API.
- Duty status indicator per day: show a tick when count_logbooks equals count_schedules, otherwise show the number of remaining duties.
- Legend below the calendar.
- Tapping a date shows a placeholder page ("Detail page coming soon"). You do not need to build the detail screen.

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Hours to Limit Section: Detailed Spec

The rolling sum math for this section runs in your NestJS service, not on the frontend. The frontend only renders what the API returns.

A. Limit Summary Cards

Four cards showing current hours flown against the regulatory limit:

Card	Limit	Sum window
Daily	8 hours	Today only
Weekly	40 hours	Rolling 7 days
Monthly	100 hours	Rolling 30 days
Annual	1050 hours	Rolling 365 days

Each card should show current hours in the window, the limit, and a visual indicator.

B. Flight Hours Trend Chart

- **X axis:** days. Show 7 days before today, today, and 7 days after today. Today is always centered.
- **Y axis:** hours. The Y max changes with the selected range toggle.
- **Red limit line** drawn horizontally at the limit value for the selected range.
- **Toggle** with options 1w, 1m, 3m, 6m, 1y. Default to 1w.

C. Chart Bounds per Toggle

Toggle	Sum window	Limit line (red)	Chart Y max
1w	Last 7 days	40 hours	45 hours
1m	Last 30 days	100 hours	125 hours
3m	Last 90 days	300 hours	325 hours
6m	Last 180 days	600 hours	625 hours
1y	Last 365 days	1050 hours	1200 hours

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D. Things to Handle

The dataset contains cases you will need to think about:

- Days with zero flight hours contribute 0 to the sum. Do not skip missing dates.
- For dates near the start of the dataset, the window can extend before the earliest date. Decide how you want to handle that.
- Future dates in the display range should still return a value. Decide what the rolling sum should mean for a future date.
- Some periods push the 1w or 1m rolling sum above the red limit line. Your chart should render those values without breaking the layout.

Design Direction

No mockups will be provided. Align with the Susi Air brand as seen on susiair.com and on Instagram at [@susiairofficial](https://www.instagram.com/susiairofficial). The style is minimalist, clean, mobile first, and operational. Think airline operations tool, not consumer travel app.

Color Palette

Role	Color
Primary navy	#0E2138
Brand red (CTAs, accents)	#E63757
Background	#F5F6F8
Card surface	#FFFFFF
Text primary	#0E2138
Text secondary	#6B7280
Success	#1FBF8F
Warning	#F59E0B
Danger	#E63757
Chart accent	#22C5E8

Typography and Components

- **Plus Jakarta Sans** from Google Fonts. Bold weights for numerical data.
- Rounded corners: 12 to 16px on cards, pill shape on primary buttons.
- Subtle card shadows. Line icons (Lucide, Phosphor, or Heroicons).
- For the Susi Air logo, use the asset from susiair.com.

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Provided Data

Three JSON files are attached. Load them into your NestJS service at startup. In memory is fine.

- `mock-flight-hours.json`: 521 days of flight hours (27 Dec 2024 to 31 May 2026) with limits and chart bounds config.
- `mock-documents.json`: pilot documents with expiry dates and threshold config.
- `mock-schedules.json`: schedule entries for April, May, and June 2026, plus the duty type legend.

Treat today as 15 May 2026 in your logic. Do not use `new Date()` to represent today. Hardcode it or make it configurable so the app behaves the same way when we review it.

Deliverables

- **A GitHub repository** with two folders (or two repositories): `/nuxt` and `/nest`.
- **Live deployments** of both apps. Vercel or Netlify for the frontend, Railway, Render, or Fly for the backend. Free tier is fine.
- **A README** covering: how to set up and run each app, required environment variables, the main choices you made and the reason behind them, and anything you would change with more time.

Timeline

Please submit within **5 calendar days** of receiving this brief. If you need more time, just let us know.

Submission

Send your GitHub repository link and both live deployment URLs to it.admin@susiair.com.